On the evolution of Maltese weak and geminate Form I verbs

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Abstract

This article charts the evolution of weak and geminate verbs in Maltese from their Proto-Arabic origins, continuing the work begun in Lucas, Spagnol & Brown (2025) on a comprehensive database of the Form I verbs in use in contemporary standard Maltese. We consider hollow, geminate, final-weak, initial-weak and irregular verbs, focusing on the evolution of the stem vowels of both the Perfect and the Imperfect, and showing how they can be accounted for on the basis of exceptionless sound laws, together with a small number of well-motivated analogical changes.

Dan l-artiklu jsegwi l-iżvilupp tal-verbi dgħajfin u torox fil-Malti min-nisel Proto-Għarbi tagħhom, u b’hekk jibni fuq ix-xogħol mibdi minn Lucas, Spagnol u Brown (2025) dwar ġabra sħiħa ta’ verbi tal-ewwel forma użati fil-Malti standard kontemporanju. Nittrattaw il-verbi moħfijin, torrox, neqsin, xebbihin u irregolari, u niffukaw fuq l-evoluzzjoni tal-vokali fiz-zokk tal-Perfett u l-Imperfett, u nuru kif jistgħu jiġu spjegati skont regoli fonetiċi bla eċċezzjonijiet, flimkien ma’ għadd żgħir ta’ bidliet analoġiċi b’motivazzjoni ċara.

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1 Introduction

In an article currently in press (Lucas, Spagnol & Brown 2025), we traced the evolution of Maltese[1] sound Form I verbs from their Proto-Arabic ancestor, focusing on regular and analogical developments in the short vowels of the verb stem. The present article

continues this work, considering weak, geminate, and irregular Form I verbs. That is, all non-augmented verbs whose roots do not consist of three distinct full consonants (e.g. *kien*, *jkun* 'to be', *ħabb*, *iħobb* 'to love').[2]

Our methodology is as in the previous article. The data collection process began with the extraction of all Form I verbs listed in Aquilina's (1987; 1990) two-volume dictionary of Maltese. The resulting list was then pruned of all obsolete and hypothetical verbs, leaving only those that are actually in use in contemporary literary or spoken standard Maltese, as attested by occurrence in the MLRS Korpus Malti, version 4.2.[3] In general, this means that they are in use in both the Perfect and Imperfect tenses, but we include a few items which today occur only in the Imperfect. Where Aquilina lists a now-obsolete Perfect for such verbs, we include it in parentheses in the tables below, as in (*ħaf*,) *iħuf* 'to prowl around'. We also consider here only verbs that are demonstrably derived from Arabic, not Romance-origin items that have been assimilated into Maltese as Form I weak verbs, such as *fada*, *jafda* 'to trust' (< Sicilian *fidari* and/or Italian *fidare*), or Maltese-internal denominal coinages, such as *dar*, *idur* 'to reside' (< *dar* 'house').

For each verb in the resulting list, we list cognates in Tunisian and Egyptian Arabic, and supply reconstructions in both the immediate common ancestor shared by Maltese, Tunisian, Egyptian and Urban Levantine Arabic—which we label Early Urban Arabic—and also in Proto-Arabic (see Lucas, Spagnol & Brown 2025 for further details on this methodology and sources for the cognates), showing how each class of weak verb has evolved from Proto-Arabic, via Early Urban Arabic, to its present-day form in Maltese.

The article is structured as follows. In Section 2 we provide the historical phonological background required for an understanding of the analyses presented in the following sections. We then deal with the various types of weak verbs in turn: hollow verbs (Section 3), geminate verbs (Section 4), final-weak verbs (Section 5), initial-weak verbs (Section 6), and finally in Section 7 we consider several highly irregular verbs. Section 8 concludes.

2 Historical phonological background

Lucas, Spagnol & Brown (2025) built on van Putten's (2020) important work establishing the sound laws linking the three short vowels of Proto-Arabic to the five of Maltese. As van Putten points out, this task is complicated by a considerable amount of subsequent analogical change, which can, however, largely be circumvented by examining only so-called primary nouns in the first instance. These are nouns which are not part of a morphologically derived paradigm. Because of this, they are much less subject to analogical influence, and consequently are a surer guide to the purely phonological changes undergone by the segments they contain.

The relevant changes, established by van Putten (2020) on the basis of primary nouns, with some minor additions provided by Lucas, Spagnol & Brown (2025), are briefly restated below. Crucial to the statement of these changes are the "guttural" and "emphatic" consonants of Arabic and earlier Maltese. The guttural consonants are defined, for the purposes of the developments under discussion, as *ʕ (Modern Standard Arabic (MSA) <ع>, Maltese <għ>), *ɣ (MSA <غ>, Maltese <għ>), *ḥ (MSA <ح>, Maltese <ħ>), *ẖ (MSA <خ>, Maltese <ħ>), and *q (MSA <ق>, Maltese <q>). The emphatic consonants are those with a secondary back articulation (dorsal or pharyngeal), namely: *ṭ (MSA <ط>, Maltese <t>), *ṣ (MSA <ص>, Maltese <s>), and *ẓ (MSA <ظ, ض>, Maltese <d>). In addition, *r (MSA <ر>, Maltese <r>) is emphatic in some but not all cases. In general, it acts as emphatic as long as it is not syllable-initial and does not follow a high front vowel. But also, all the sedentary North African varieties of Arabic to which Maltese is most closely related exhibit a phonemic split between emphatic /ṛ/ and non-emphatic /r/, such that *r in certain roots, such as √RʔS 'head', is emphatic no matter the phonological context.

In the past few centuries, Maltese has lost the distinction between emphatic and non-emphatic consonants, and merged or lost several of the guttural consonants. Emphatic and guttural contexts are therefore historical—not synchronic—conditioning environments, but the synchronic distribution of the various reflexes of the vowels of Proto-Arabic can only be understood with reference to these historical environments. This distribution is as follows.

In all contexts relevant to the stem vowels of Form I verbs, Proto-Arabic *_u_ becomes Maltese /o/ (IPA /ɔ/). In general, *_i_ remains /i/ (IPA /ɪ/), including in emphatic environments. Adjacent to a guttural consonant or *_h_ (MSA <ه>, Maltese <h>), *_i_ becomes /e/ (IPA /ɛ/). Additionally, *_i_ becomes /e/ in a post-stress closed syllable, unless that syllable contains an emphatic or the final consonant is guttural, in which case it becomes /ɑ/ (IPA /ɐ/). Proto-Arabic *_a_ becomes Maltese /e/, except in the environment of a guttural or emphatic consonant, where it remains /ɑ/.

Van Putten (2020) mentions the long vowels in passing. Since they are of particular relevance for the weak verbs, we need to address them in detail here. Two of the three Proto-Arabic long vowels are straightforward: as Cohen (1966: 19; cited by van Putten 2020: 64) notes, *_ū_ and *_ī_ are reflected as Maltese /ū/ (IPA /uː/) and /ī/ (IPA /iː/) respectively. The outcome of Proto-Arabic *_ā_ is more complex, as it splits into two distinct Maltese phonemes: /ɑ̄/ (IPA /ɐː/) and /ie/ (IPA /ɪː/). Many Arabic varieties have a raised and fronted allophone of *_ā_, appearing in complementary distribution with a low allophone. The latter is restricted to “back” consonantal environments (the precise definition of which varies from dialect to dialect), while the former appears elsewhere. This raising phenomenon is known in the Arabic grammatical tradition as *_imāla_*. In the case of the Maltese *_imāla_, we are dealing with a phonemic split, as the aforementioned losses and mergers of back consonants mean that the environments in which the two reflexes of Proto-Arabic *_ā_ appear are now overlapping rather than complementary.

As we will see, the distribution of these two reflexes of *_ā_ in Maltese has been significantly impacted by analogical changes in the Perfect tense of hollow verbs (as well as in a number of other morphologically derived forms, such as active participles, which are not our focus here). In order to understand these analogical changes, it is necessary to first establish the precise environment in which *_imāla_ was blocked in earlier Maltese, prior to the consonantal changes which disrupted that environment.

Borg (1976) discusses the issue of *_imāla_ in Maltese and Arabic varieties in some detail. His point of departure is the observation that one can capture a good part of the data with a rule stating that the low reflex of *_ā_ “is maintained [in Maltese] in the neighbourhood of a segment that is historically ‘plus back’” (1976: 203). He then goes on to point out numerous exceptions to this rule, including several alternate forms in the Form I hollow verbs (such as *_diem~dam_ ‘to linger’ and *_sieb~sab_ ‘to find’), which we will consider in Section 3. His suggestion for how to explain these exceptions is that at “some later stage in its history, Maltese… came under the influence of an *_imaala_-less

dialect" (Borg 1976: 220). But before we resort to such speculations, we should first make sure that our initial statement of the blocking environment for *imāla* is correct. Borg (1976: 191) explains that his "plus back" encompasses all of the emphatic and guttural consonants listed above. But we should not expect a priori that *imāla* will be sensitive to the presence of guttural consonants. Borg (1976: 206) himself points out that there are some Arabic varieties in which gutturals only block *imāla* when they precede **ā*, and others in which a guttural environment does not block *imāla* at all.

In order to discover where Maltese fits into this typology, we need to follow van Putten's (2020) lead and examine primary nouns. Several examples of these show that *imāla* is blocked in what was historically an emphatic environment, such as *ras* 'head' (< **ṛās*) and *dar* 'house' (< **dāṛ*). The only putative counterexample is *tieqa* 'window', which appears to derive from **ṭāqa*. However, in the North African dialects of Arabic to which Maltese is most closely related, this word is typically realized with a non-emphatic initial consonant,[4] this being an instance of a more general rule in these dialects of root consonants losing emphasis before a following uvular (cf. Souag 2013: 37). The source of Maltese *tieqa* is therefore **tāqa*, and *tieqa* is thus not a counterexample to the rule that *imāla* is blocked in an emphatic environment. Instead, what this item shows is that *imāla* is not blocked when a guttural follows. Further examples of this are *sieq* 'leg' (< **sāq*) and *siegħa* 'hour' (< **sāʕa*). Van Putten (2020: 64) suggests that *imāla* is blocked when a guttural precedes, but fails to give a primary-noun example, giving only the verb *qal* 'to say' (< **qāl*). Primary nouns in which a guttural precedes **ā* without a following emphatic are very scarce, but there is one clear example: *qiegħ* 'bottom' (< **qāʕ*).

The data therefore indicates that *imāla* was blocked in Maltese simply in the environment of emphatic consonants, before these were lost, and not in the environment of gutturals. In the following section, we will see this manifested in the hollow verbs, where exceptions are shown to result from a relatively simple operation of analogy.

3 Hollow verbs

The hollow verbs in Maltese and Arabic varieties are those which derive from Proto-Arabic roots in which the medial radical (R2) is a glide: *w* or *y*. It seems that in Proto-Arabic roots with glides in medial and final position behaved regularly, at least in the Perfect of Form I verbs (van Putten 2017). Thus, just as ordinary Perfects were of three kinds, depending on the vowel after the second radical (*CɑCɑC, *CɑCiC and *CɑCuC), verbs with a medial glide also exhibited the same three classes: *CɑWɑC, *CɑWiC and *CɑWuC (where W stands for *w* or *y*). In Early Urban Arabic, however, as well as in the variety of Classical Arabic which became MSA, these medial triphthongs were all monophthongized as *ā*, giving us the characteristic Perfect hollow verb pattern: *CāC.

Form I verbs in Proto-Arabic also vary with respect to the vowel after the second radical and the vowel on the prefix in the Imperfect. The basic pattern here is *yVCCVC and there are three specific arrangements of vowels that occur: *yiCCɑC, *yɑCCiC and *yɑCCuC.[5] In the hollow verbs, these Imperfect patterns collapse to *yiCāC, *yɑCīC and *yɑCūC already in Proto-Arabic (cf. Schuchardt 2016). These are then carried into Early Urban Arabic as *yiCāC, *yɑCīC and *yiCūC (the latter with optional vowel harmony, giving *yuCūC). Of these three, *yiCāC is the least common. The vast majority of hollow verbs have either **ī* or **ū* in the Imperfect, and in all of these, **y* as R2 is associated with **ī* in the Imperfect and **w* as R2 is associated with **ū* (e.g. √KWN: **kān*, *y(i/u)kūn* 'to be', √ZYD: **zād*, *yazīd* 'to increase'). In contemporary Arabic dialects, the yiCāC Imperfect pattern is generally restricted to three verbs: *ḫāf*, *yiḫāf* 'to fear', *bāt*, *yibāt* 'to spend the night', and *nām*, *yinām* 'to sleep'. Maltese has lost all of these (*ħaf*, *iħuf* 'to prowl around' derives from Proto-Arabic **ḥāf*, *yaḥūf* 'to be next to'). All Maltese hollow verbs therefore have either /ū/ or /ī/ in the Imperfect, having retained the original medial vowel of the Imperfect in every case.

The medial vowel in the Perfect of Maltese hollow verbs is always a reflex of **ā*. In most cases it is the expected reflex: /ā/ when R1 or R3 was originally emphatic (as in Table 1) and /ie/ when neither are emphatic (as in Table 2).

Table 1: Regular hollow verbs with emphatic roots

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
dar, *idur* 'to turn'	**dār*, *y(i/u)dūr*	**dawar*, *yadūr*	*dār*, *ydūr*	*dār*, *yudūr*[6]
far, *ifur* 'to overflow'	**fār*, *y(i/u)fūr*	**fawar*, *yafūr*	—	*fār*, *yufūr*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
sam, *isum* ‘to fast’	**ṣām*, *y(i/u)ṣūm*	**ṣawam*, *yaṣūm*	*ṣām*, *yṣūm*	*ṣām*, *yuṣūm*
sar, *isir* ‘to become’	**ṣār*, *yaṣīr*	**ṣayar*, *yaṣīr*	*ṣār*, *yṣīr*	*ṣār*, *yiṣīr*
tal, *itul* ‘to be(come) long’	**ṭāl*, *y(i/u)ṭūl*	**ṭawul*, *yaṭūl*	*ṭāl*, *yṭūl*	*ṭāl*, *yuṭūl*
tar, *itir* ‘to fly’	**ṭār*, *yaṭīr*	**ṭawar*, *yaṭīr*	*ṭār*, *yṭīr*	*ṭār*, *yiṭīr*
żar, *iżur* ‘to visit’	**zār*, *y(i/u)zūr*	**zawar*, *yazūr*	*zār*, *yzūr*	*zār*, *yuzūr*

Table 2: Regular hollow verbs with non-emphatic roots

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
biel, *ibul* ‘to urinate’	**bāl*, *y(i/u)būl*	**bawal*, *yabūl*	*bāl*, *ybūl*	*bāl*, *yubūl*
bies, *ibus* ‘to kiss’	**bās*, *y(i/u)būs*	**bawas*, *yabūs*	*bās*, *ybūs*	*bās*, *yubūs*
(*fieħ*,) *ifuħ* ‘to smell sweet’	**fāḥ*, *y(i/u)fūḥ*	**fawaḥ*, *yafūḥ*	*fāḥ*, *yfūḥ*	*fāḥ*, *yufūḥ*
fieq, *ifiq* ‘to recover’	**fāq*, *yafīq*/*y(i/u)fūq*	**ʔafwaq*, *yufīq* / **fawaq*, *yafūq*[7]	*fāq*, *yfīq*	*fāʔ*, *yufūʔ*
kien, *ikun* ‘to be’	**kān*, *y(i/u)kūn*	**kawun*, *yakūn*	*kān*, *ykūn*	*kān*, *yukūn*
(*liem*,) *ilum* ‘to scold’	**lām*, *y(i/u)lūm*	**lawam*, *yalūm*	*lām*, *ylūm*	*lām*, *yulūm*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
miel, *imil* ‘to lean’	**māl*, *yamīl*	**mayal*, *yamīl*	*māl*, *ymīl*	*māl*, *yimīl*
miet, *imut* ‘to die’	**māt*, *y(i/u)mūt*	**mawit*, *yamūt*	*māt*, *ymūt*	*māt*, *yumūt*
niek, *inik* ‘to spoil’	**nāk*, *yanīk*	**nayak*, *yanīk*	*nāk*, *ynīk*	*nāk*, *yinīk*
qiem, *iqim* ‘to venerate’	**qām*, *yaqīm*	**ʔaqwam*, *yuqīm*	*qām*, *yqīm*	*ʔām*, *yiʔīm*
qies, *iqis* ‘to measure’	**qās*, *yaqīs*	**qayas*, *yaqīs*	*qās*, *yqīs*	*ʔās*, *yiʔīs*
ried, *irid* ‘to want’	**rād*, *yarīd*	**ʔarwad*, *yurīd*	—	*rād*, *yirīd*
sies, *isus* ‘to insist’	**sās*, *y(i/u)sūs*	**sawas*, *yasūs*	—	—
żied, *iżid* ‘to increase’	**zād*, *yazīd*	**zayad*, *yazīd*	*zād*, *yzīd*	*zād*, *yizīd*

However, there are several cases where we do not get the expected reflex. That is, in these cases we have, at least optionally, /ie/ where /ā/ would be expected, and vice versa. These are presented in Table 3 and Table 4 respectively.

Table 3: Hollow verbs with /ie/ despite emphatic radicals

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
(*bad/*)*bied*, *ibid* ‘to lay eggs’	**bāðˤ*, *yabīðˤ*	**bayaɮˤ*, *yabīɮˤ*	*bāðˤ*, *ybīðˤ*	*bāḍ*, *yibīḍ*
(*ħat/*)*ħiet*, *iħit* ‘to sew’	**ḫāṭ*, *yaḫīṭ*	**ḫayaṭ*, *yaḫīṭ*	—	—
sab/sieb, *isib* ‘to find’	**ṣāb*, *yaṣīb*	**ṣayab*, *yaṣīb*	—	*ṣāb*, *yiṣīb*

Table 4: Hollow verbs with /ā/ but no emphatic radicals

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
dieb/*dab*, *idub* 'to melt'	**ðāb*, *y(i/u)ðūb*	**ðawab*, *yaðūb*	*ðāb*, *yðūb*	*dāb*, *yudūb*
diem/*dam*, *idum* 'to continue'	**dām*, *y(i/u)dūm*	**dawam*, *yadūm*	*dām*, *ydūm*	*dām*, *yudūm*
(*dieq*/)*daq*, *iduq* 'to taste'	**ðāq*, *y(i/u)ðūq*	**ðawaq*, *yaðūq*	*ðāq*, *yðūq*	*dāʔ*, *yudūʔ*
ġieb/*ġab*, *iġib* 'to bring'	**gāb*, *yagīb*	**gayaʔ bi*, *yagīʔ bi*	*žāb*, *yžīb*	*gāb*, *yigīb*
għam, *jgħum* 'to swim'	**ʕām*, *y(i/u)ʕūm*	**ʕawam*, *yaʕūm*	*ʕām*, *yʕūm*	*ʕām*, *yuʕūm*
(*ħaf*,) *iħuf* 'to prowl about'	**ḥāf*, *y(i/u)ḥūf*	**ḥawaf*, *yaḥūf*	—	—
qam, *iqum* 'to get up'	**qām*, *y(i/u)qūm*	**qawam*, *yaqūm*	*qām*, *yqūm*	*ʔām*, *yuʔūm*
saq, *isuq* 'to drive'	**sāq*, *y(i/u)sūq*	**sawaq*, *yasūq*	—	*sāʔ*, *yusūʔ*
żal, *iżul* 'to evade'	**zāl*, *y(i/u)zūl*	**zawal*, *yazūl*	—	*zāl*, *yuzūl*

To explain these anomalies, it is not necessary to invoke contact with an *imāla*-less dialect. Instead, we can explain them in terms of an analogical change of a kind already well-established in Arabic varieties. As we pointed out in Lucas, Spagnol & Brown (2025) regarding Imperfect strong verbs, in the transition from Proto-Arabic to Early Urban Arabic, there came to be a strong (but not exceptionless) association between an emphatic R2 or R3 and a /u/ as the stem vowel (e.g. **yafṭim* > **y(i/u)fṭum* 'to wean'), and a non-emphatic R2 or R3 and /i/ as the stem vowel (e.g. **yaktub* > **yaktib* 'to write'). Maltese inherited a similar, though weaker, association in hollow

verbs between emphatic radicals and a /ū/ stem vowel in the Imperfect. This can be seen in Table 1, where five of the seven hollow verbs with an emphatic R1 or R3 have /ū/ as the stem vowel in the Imperfect.

Now consider again the three verbs in Table 3. These are verbs that at least optionally have /ie/ in the Perfect, when we would expect /ā/, given the emphatic radicals. Note that all of these verbs have /ī/ as the stem vowel in the Imperfect. Conversely, regarding the verbs in Table 4, which anomalously have /ā/ in the Imperfect when we would expect /ie/, eight out of nine have /ū/ as the stem vowel in the Imperfect. The obvious conclusion is that there has been an analogical extension, within the history of Maltese proper, of the already existing association between /ū/ in the Imperfect and /ā/ in the Perfect, and therefore also between /ī/ in the Imperfect and /ie/ in the Perfect.

Note that analogical change, by its nature, works via tendencies, optionality, and semi-regularity. Thus we have several verbs that have resisted the pressure of analogy, such as *sar*, *isir* 'to become' (< **ṣār*, *yaṣīr*), which has not developed a Perfect with /ie/, or *kien*, *ikun* 'to be', which has not developed a Perfect with /ā/. There is also the isolated case of *ġieb*, *iġib* 'to bring', which has developed the alternative Perfect form *ġab* in the absence of a clear analogical model for doing so.

Exceptions of this kind are not unexpected and do not undermine the basic conclusion: the phonology of a complex system, such as hollow verbs, does not evolve at random, but rather through the interaction of regular sound change and semi-regular analogy, both of which can be straightforwardly reconstructed. In Section 4 we apply the same approach to geminate verbs.

Before that, there is one more small class of hollow verbs to consider. These are the four verbs shown in Table 5, which begin with <għ> (< *ʕ < [*ʕ,*ɣ]) and have /ē/ as the stem vowel in the Perfect.

Table 5: <*għ*>-initial hollow verbs with /ē/

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
għeb, *jgħib* 'to disappear'	**ɣāb*, *yaɣīb*	**ɣayab*, *yaɣīb*	*ɣāb*, *yɣīb*	*ɣāb*, *yiɣīb*
għen, *jgħin* 'to help'	**ʕān*, *yaʕīn*	**ʕawan*, *yaʕīn*	*ʕān*, *yʕīn*	*ʕān*, *yiʕīn*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
għer, *jgħir* ‘to envy’	**ɣār*, *yaɣīr*/*yiɣār*[8]	**ɣayar*, *yaɣīr* / **ɣayir*, *yiɣār*	*ɣār*, *yɣīr*	*ɣār*, *yiɣīr*
għex, *jgħix* ‘to live’	**ʕāš*, *yaʕīš*	**ʕayaš*, *yaʕīš*	*ʕāš*, *yʕīš*	*ʕāš*, *yiʕīš*

Three of these verbs contain no emphatic consonant, so *imāla* is expected. In *għer* (< **ɣār*) the final /r/ should block *imāla*. The fact that we have it is to be explained, as with the verbs in Table 3, as an extension of the association between /ī/ in the Imperfect and *imāla* in the Perfect. The fact that, in the case of the verbs in Table 3, *imāla* has resulted in /e/ rather than /ie/ is evidently conditioned by the initial pharyngeal. This is not unexpected. Cairene Arabic is an example of a related Arabic variety in which only emphatics block *imāla*,[9] but where, in the environment of a pharyngeal, /ā/ can optionally be realized with a different allophone ([aː]) to the one found in other *imāla* environments ([ɛ̈ː]) (Woidich 2006: 9). Maltese must have inherited a similar system of allophony, which later led to a phonemic split between /ē/ and /ie/, alongside the split between /ā/ and /ie/.

4 Geminate verbs

Geminate verbs are those in which R2 and R3 are identical. As Huehnergard (2005: 171–6) shows, in Proto-Semitic (as reflected in Akkadian and Ge’ez), these verbs behaved just like ordinary strong verbs. But in Proto-Central-Semitic, and therefore Proto-Arabic, there is a sound change such that a short vowel between R2 and R3 is deleted whenever a vowel (which could be any inflectional suffix) follows R3. Thus we have, for example, Proto-Arabic **ɮʕaraba*, **yaɮʕribu* ‘to hit’, but **dalla*, *yadullu* ‘to indicate’, rather than ***dalala*, *yadlulu*.[10]

As noted in Section 3, there are three patterns in the Perfect of Proto-Arabic sound verbs: *CɑCɑC, *CɑCiC and *CɑCuC. In the Proto-Arabic geminate verbs the second vowel is simply deleted in the third person, and we have a single pattern *CɑCC, with /

ɑ/. As explained in Section 2, short **a* becomes /e/ (IPA /ɛ/) in Maltese, except in the environment of an emphatic or guttural consonant, where it remains /ɑ/ (IPA /ɐ/). It is therefore as expected that we find two patterns in the Perfect of Maltese geminate verbs: CɑCC and CeCC, with CɑCC occurring mostly when R1 or R2/3 is guttural or emphatic, and CeCC mostly otherwise. Table 6 shows all instances of CɑCC verbs in Maltese where the /ɑ/ is the regular outcome of **a* adjacent to a guttural or emphatic, while Tables 7 and 8 show all instances of CeCC verbs where the /e/ in the Perfect is the regular outcome of **a* in the absence of a guttural or emphatic.

Table 6: Regular geminate verbs with emphatic or guttural roots

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
bass, *iboss* ‘to fart’	**baṣṣ*, *y(i/u)buṣṣ*	**baṣṣ*, *yabuṣṣ*	*baṣṣ*, *ybuṣṣ*	—
damm, *idomm* ‘to collect’	**ðamm*, *y(i/u)ðumm*	**ʒˤamm*, *yaʒˤumm*	—	*ḍamm*, *yuḍumm*
daqq, *idoqq* ‘to play an instrument’	**daqq*, *y(i/u)duqq*	**daqq*, *yaduqq*	*daqq*, *yduqq*	*daʔʔ*, *yuduʔʔ*
ġarr, *iġorr* ‘to carry’	**garr*, *y(i/u)gurr*	**garr*, *yagurr*	—	*garr*, *yugurr*
ħabb, *iħobb* ‘to love’	**ḥabb*, *y(i/u)ḥubb/yaḥibb*	**ḥabb*, *yaḥubb/yiḥabb*	*ḥabb*, *yḥibb*	*ḥabb*, *yiḥibb*
ħakk, *iħokk* ‘to scratch’	**ḥakk*, *y(i/u)ḥukk*	**ḥakk*, *yaḥukk*	*ḥakk*, *yḥukk*	*ḥakk*, *yuḥukk*
ħall, *iħoll* ‘to loosen’	**ḥall*, *y(i/u)ḥull/yaḥill*	**ḥall*, *yaḥull*	*ḥall*, *yḥill*	*ḥall*, *yiḥill*
ħass, *iħoss* ‘to feel’	**ḥass* *y(i/u)ḥuss/yaḥiss*	**ḥass*, *yaḥuss/yaḥiss*	*ḥass*, *yḥiss*	*ḥass*, *yiḥiss*
ħatt, *iħott* ‘to unload’	**ḥaṭṭ*, *y(i/u)ḥuṭṭ*	**ḥaṭṭ*, *yaḥuṭṭ*	*ḥaṭṭ*, *yḥuṭṭ*	*ḥaṭṭ*, *yuḥuṭṭ*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
ħażż, *iħożż* ‘to note’	**ḥazz*, *y(i/u)ḥuzz/yaḥizz*	**ḥazz*, *yaḥuzz*	*ḥazz*, *yḥizz*	*ḥazz*, *yiḥizz*
radd, *irodd* ‘to return’	**radd*, *y(i/u)rudd*	**radd*, *yarudd*	*ṛadd*, *yṛudd*[11]	*radd*, *yurudd*
rass, *iross* ‘to press’	**raṣṣ*, *y(i/u)ruṣṣ*	**raṣṣ*, *yaruṣṣ*	—	*raṣṣ*, *yuruṣṣ*
raxx, *iroxx* ‘to sprinkle’	**rašš*, *y(i/u)rušš*	**rašš*, *yarušš*	*ṛašš*, *yṛušš*	*rašš*, *yurušš*
sarr, *isorr* ‘to pack up’	**ṣarr*, *y(i/u)ṣurr*	**ṣarr*, *yaṣurr*	*ṣarr*, *yṣurr*	*ṣarr*, *yuṣurr*
żaqq, *iżoqq* ‘to feed (birds)’	**zaqq*, *y(i/u)zuqq*	**zaqq*, *yazuqq*	—	*za??*, *yuzu??*

Table 7: Regular geminate verbs with non-emphatic and non-guttural roots

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
bell, *ibill* ‘to dip’	**ball*, *yabill*	**ball*, *yabull*	*ball*, *ybill*	*ball*, *yibill*
kedd, *ikidd* ‘to vex’	**kadd*, *yakidd*	**kadd*, *yakudd*	—	*kadd*, *yikidd*
ġeżż, *iġiżż* ‘to shear’	**gazz*, *yagizz*	**gazz*, *yaguzz*	*zazz*, *yzizz*	*gazz*, *yigizz*
medd, *imidd* ‘to spread’	**madd*, *yamidd*	**madd*, *yamudd*	*madd*, *ymidd*	*madd*, *yimidd*
senn, *isinn* ‘to whet’	**sann*, *yasinn*	**sann*, *yasunn*	—	*sann*, *yisinn*

Table 8: Geminate verbs with regular /e/ in the Perfect but irregular /e/ in the Imperfect

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
kenn, *ikenn* 'to shelter'	**kann*, *yakinn*	**kann*, *yakinn*	—	*kann*, *yikinn*
temm, *itemm* 'to finish'	**tamm*, *yatimm*	**tamm*, *yatimm*	*tamm*, *ytimm*	*tamm*, *yitimm*
xedd, *ixidd*/*ixedd* 'to put on'	**šadd*, *yašidd*	**šadd*, *yašidd*	*šadd*, *yšidd*	*šadd*, *yišidd*

Corresponding to the Imperfect patterns of Proto-Arabic sound verbs (*yaCCiC, *yaCCuC and *yiCCaC), there are three patterns for the Imperfect of geminate verbs: *yaCiCC, *yaCuCC and *yiCaCC. Just as *yiCāC is the least common Imperfect pattern in hollow verbs, *yiCaCC is the least common geminate pattern. No verbs with this pattern survive as such into contemporary standard Maltese (but note **ḥabb* 'to love' in Table 7 and **saff* 'to suck' in Table 10). The most frequent Imperfect pattern in Maltese geminate verbs is iCCoC. This is the regularly expected reflex of *yaCCuC. Note in particular that every verb in Table 6 has a regularly inherited Imperfect of this pattern, while all the regularly developed verbs in Table 7 have iCCiC Imperfects, the regular reflex of *yaCCiC in the absence of a guttural or emphatic consonant.

Table 8 shows verbs which have /e/ as the regularly expected reflex of **a* in the Perfect (due to the lack of gutturals or emphatics), but which also have /e/ in the Imperfect, where the regularly expected reflex of **i* here is /i/, as in Table 7. This irregular /e/ in the Imperfect is best understood as levelling or harmonization of the stem vowels within individual verb paradigms. Consider *xedd* 'to put on (clothes)'. This has two variant Imperfect forms: the regularly expected *ixidd*, and an innovative form *ixedd*. Presumably there used to be similar variation in the Imperfects of the other verbs in Table 8 before the original forms with /i/ were lost.

We also see this tendency towards harmonization between Perfect and Imperfect in the verbs in Table 9, except here the /e/ is the regular outcome of **i* adjacent to a guttural in the Imperfect, but irregular in the Perfect, since the presence of gutturals

would normally cause **a* to be reflected as /ɑ/, as in Table 6. Consider in Table 9 the doublets *qarr*/*qerr*, *iqerr* 'to confess' and *raqq*/*reqq*, *ireqq* 'to get thin'. Here we have the regularly inherited forms with /ɑ/ preserved alongside the innovative forms with /e/, harmonizing with the regularly inherited /e/ in the Imperfects. As before, we can assume that similar doublets existed for the other verbs in this table, before the original forms were lost.

Table 9: Geminate verbs with /e/ in the Perfect despite presence of guttural or emphatic

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
ħebb, *iħebb*[12] 'to attack'	**habb*, *yahibb*	**habb*, *yahubb*	*habb*, *yhibb*	*habb*, *yihibb*
ħenn, *iħenn* 'to have mercy'	**ḥann*, *yaḥinn*	**ḥann*, *yaḥinn*	—	*ḥann*, *yiḥinn*
leħħ, *ileħħ* 'to flash'	**laḥḥ*, *yaliḥḥ*	**ʔalaḥḥ*, *yuliḥḥ*	—	*laḥḥ*, *yiliḥḥ*
qarr/*qerr*, *iqerr* 'to confess'	**qarr*, *yaqirr*	**ʔaqarr*, *yuqirr*	*qarr*, *yqirr*	*ʔarr*, *yiʔirr*
raqq/*reqq*, *ireqq* 'to get thin'	**raqq*, *yariqq*	**raqq*, *yariqq*	—	*raʔʔ*, *yiriʔʔ*

Finally, we have the verbs in Table 10, which have /ɑ/ in the Perfect where /e/ is expected, since there are no gutturals or emphatics. A particularly instructive verb here is 'to whet'. This is **sann* in Early Urban Arabic, with two possible Imperfects: **y(i/u)sunn* with stem vowel /u/, continuing Proto-Arabic **yasunn*, and also innovative **yasinn*, exhibiting the tendency, described in Section 3, to replace an original stem vowel /u/ with /i/ in non-emphatic roots, as in **yaktub* > **yaktib* (and as also evident with the verbs in Table 7). Maltese has inherited both of these Imperfects as *isonn* and *isinn* respectively. Thus, one regularly expected paradigm for this verb, given the lack of emphatics or gutturals, would be *senn*, *isinn*. This is indeed attested, and we have

therefore listed it among the regular verbs in Table 7. The form *isonn* is also regularly expected, but the alternative attested Perfect form *sann*, listed in Table 10, is not. This form with /ɑ/, along with all the other irregular /ɑ/ Perfects in Table 10, is best explained in directly parallel fashion to the process we saw in Section 3 regarding hollow verbs. There we saw an analogical extension of the association between /ū/ in the Imperfect and /ā/ in the Perfect to verbs in which the regularly expected reflex of **ā* is /ie/, given the lack of emphatic R1 or R3. With the geminate verbs, we have an even stronger inherited association between /o/ in the Imperfect and /ɑ/ in the Perfect: note that every verb in Table 6 has a regularly inherited /o/ in the Imperfect. In Table 10, where all the Maltese verbs have regularly inherited /o/ stem vowels in the Imperfect, we see the extension of this /o/–/ɑ/ association to verbs which originally must have had /e/ as their stem vowel in the Perfect.

Table 10: Geminate verbs with /ɑ/ in the Perfect despite no guttural or emphatic

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
sadd, *isodd* ‘to block’	**šadd*, *y(i/u)sudd* / *yasidd*	**sadd*, *yasudd*	*sadd*, *ysidd*	*sadd*, *yisidd*
(*seff*/*saff*), *isoff*[13] ‘to suck’	**saff*, *y(i/u)suff*/*yasiff*	**saff*, *yisaff*	—	*saff*, *yisiff*
sann, *isonn* ‘to whet’	**sann*, *y(i/u)sunn* / *yasinn*	**sann*, *yasunn*	—	*sann*, *yisinn*
xamm, *ixomm* ‘to smell’	**šamm*, *y(i/u)šumm* / *yašimm*	**šamm*, *yašumm*	*šamm*, *yšimm*	*šamm*, *yišimm*
żamm, *iżomm* ‘to keep’	**zamm*, *y(i/u)zumm* / *yazimm*	**zamm*, *yazumm*	—	*zamm*, *yizimm*

Thus we see here, as with the hollow verbs discussed in Section 3, that an apparently complex dataset can be accounted for by means of regular sound laws and the operation of a limited number of straightforward analogical tendencies. In the next

section we consider the final-weak verbs, where we once again observe similar processes at work.

5 Final-weak verbs

Final-weak verbs are those in which R3 is **w* or **y*. The evidence of Safaitic inscriptions and the Quranic consonantal text suggests that such roots behaved regularly in Proto-Arabic (van Putten 2017; 2020: 26–29), except that /iw/ and /uy/ sequences were likely realized as **ī* (or **iy*) and **ū* (or **uw*), respectively, already in Proto-Semitic (Huehnergard 2019: 52). However, in all Arabic dialects and in the variety of Classical Arabic that becomes MSA, the inherited final triphthong **VWV* becomes a monophthong (long in MSA, generally short in the dialects, where vowel length is not distinctive in unstressed word-final open syllables). Thus for Proto-Arabic we can reconstruct the following Perfect patterns: *CɑCɑW(ɑ), *CɑCuW(ɑ), *CɑCiW(ɑ). In MSA, these are reflected as CɑCā, CɑCā, and CɑCiy(ɑ), respectively, and in Early Urban Arabic they can be reconstructed as *CɑCɑ, *CɑCɑ, and *CiCi, respectively. The Proto-Arabic Imperfect patterns can be reconstructed as *yɑCCiW(V), *yɑCCuW(V), and *yiCCɑW(V). In MSA, these are reflected as yɑCCī, yɑCCū, and yɑCCā, respectively, and in Early Urban Arabic they can be reconstructed as **yaCCi*, **y(i/u)CCu*, and **yiCCa*, respectively. In the dialects which descend from Early Urban Arabic, however, /u/-final Imperfects are rare, generally being levelled to /i/-final Imperfects (as in MSA *yadʕū* 'to call' versus Cairene *yidʕi*, Maltese *jidgħi*), occasionally also to /ɑ/-final (as in MSA *yadhū* 'to afflict' versus Cairene *yidhi*, Maltese *jedha* 'to kill time').[14]

At an early stage in the history of the Maghrebi dialect group, to which Maltese and Tunisian belong, there was considerable levelling in the Perfect of final-weak verbs, such that now all original *CiCi (< *CɑCiW) Perfects have been levelled to *CɑCɑ. These are reflected in Maltese as either CɑCɑ Perfects, all of which have the expected guttural or emphatic as R1 or R2 (see Table 11), or CeCɑ Perfects, most of which lack gutturals or emphatics (see Table 12). All CeCɑ verbs that do contain gutturals or emphatics (Table 13) are straightforwardly accounted for by means of an analogical process familiar from the preceding sections.

Unlike the hollow and geminate verbs considered in the previous sections, R1 and R2 form a consonant cluster in the Imperfect of final-weak verbs. This means that, in Maltese, prefix vowels are maintained in these verbs, as they are in sound verbs. In Lucas, Spagnol & Brown (2025), we discussed the so-called Barth-Ginsberg alternation (cf. footnote 4), which concerns these prefix vowels and their interaction with stem vowels. Specifically, in Proto-Arabic, the prefix vowel is /i/ in verbs with *CɑCiC Perfects, and otherwise the prefix vowel is /ɑ/. This alternation is maintained in Early Urban Arabic, except that when the stem vowel is /u/, the prefix vowel varies optionally between /i/ and /u/. Most Arabic dialects level the prefix vowel to /i/ (or /u/) across the board. In Maltese and Tunisian, however, **a* as the inherited prefix vowel is maintained in strong verbs when R1 is guttural, and sporadically maintained when R1 is /r/, and otherwise levelled to /i/. In both Maltese and Tunisian (Singer 1984: 360) final-weak verbs, we see a similar alternation between /ɑ/ and /i/, but here there is no longer any connection with the Perfect: /ɑ/ is now the prefix vowel simply whenever R1 is guttural (and sometimes when it is /r/). Otherwise, it is /i/ (or rarely /e/ or /o/ in Maltese).

The Maltese verbs that have developed regularly according to the processes described above are presented in Tables 11 and 12.

Table 11: Regular final-weak verbs with emphatic or guttural R1 or R2

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
baqa', jibqa' 'to stay'	**biqi, yibqa*	**baqiy, yibqay*	*bqa, yibqa*	*biʔi, yibqaʿ*
dagħa, jidgħi 'to blaspheme'	**daʕa, y(i/u)dʕu*	**daʕaw, yadʕuw*	*dʕa, yidʕi*	*daʕa, yidʕi*
dara, jidra 'to get used to'	**ðiri, yiðra*	**ɮʕariy, yiɮʕraw*	—	—
ġara, jiġri 'to happen'	**gara, yagri/yigra*	**garay, yagriy*	*žṛa, yižṛa*	*gara, yigra*
għama, jagħma 'to go blind'	**ʕimi, yaʕma*	**ʕamiy, yaʕmay*	—	*ʕimi, yiʕmi*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
għama, *jagħmi* ‘to blind’	**ʕama*, *yaʕmi*	**ʔaʕmay*, *yuʕmiy*	—	*ʕama*, *yiʕmi*
għana, *jagħni* ‘to enrich’	**ɣana*, *yaɣni*	**ʔaɣnay*, *yuɣniy*	*ɣna*, *yaɣni*	*ɣana*, *yiɣni*
għawa, *jagħwi* ‘to howl	**ʕawa*, *yaʕwi*	**ʕaway*, *yaʕwiy*	—	*ʕawa*, *yiʕwi*
ħara, *jaħra* ‘to defecate’	**ḫiri*, *yiḫra*	**ḫariʔ*, *yiḫraʔ*	—	*ḫiri*, *yiḫri*
(*ħata*,) *jaħti* ‘to be guilty’	**ḫaṭa*, *yaḫṭi* / *ḫiṭi*, *yiḫṭa*	**ʔaḫṭaʔ*, *yuḫṭiʔ* / *ḫaṭiʔ*, *yiḫṭaʔ*	—	*ḫaṭa*, *yiḫṭi*
qara, *jaqra* ‘to read’	**qara*, *yaqra*	**qaraʔ*, *yaqraʔ*	*qra*, *yaqra*	*ʔara*, *yiʔra*
ragħa, *jirgħa* ‘to pasture’	**raʕa*, *yarʕa*	**raʕay*, *yarʕay*	—	*raʕa*, *yirʕa*
ragħa, *jirgħi* ‘to froth’	**raɣa*, *yarɣu*	**raɣaw*, *yarɣuw*	—	*raɣa*, *yirɣi*
rama, *jarmi* ‘to throw’	**rama*, *yarmi*	**ramay*, *yarmiy*	*ṛma*, *yaṛma*	*rama*, *yirmi*

Table 12: Regular final-weak verbs with non-emphatic, non guttural R1 and R2

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
beka, *jibki* ‘to cry’	**bakal biki*, *yabki*	**bakay*, *yabkiy*	*bka*, *yibki*	*baka*, *yibki*
bena, *jibni* ‘to build’	**bana*, *yabni*	**banay*, *yabniy*	*bna*, *yibni*	*bana*, *yibni*
deha, *jedha* ‘to kill time’	**daha*, *yadhu*	**dahaw*, *yadhuw*	—	*daha*, *yidhi*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
dewa, *jidwi* ‘to echo’	**dawa*, *yadwi*	**daway*, *yadwiy*	*dwa*, *yidwi*	—, *yidwi*
feda, *jifdi* ‘to redeem’	**fada*, *yafdi*	**faday*, *yafdiy*	*fda*, *yifdi*	*fada*, *yifdi*
fela, *jifli* ‘to inspect’	**fala*, *yafli*/*y(i/u)flu*	**falaw*, *yafluw*	*fla*, *yifli*	—
fena, *jifni* ‘to waste away’	**fini*/*fana*, *yafni*/*yifna*	**faniy*, *yifnay*	—	*fini*, *yifna*
fesa, *jifsa* ‘to fart’	**fasa*, *y(i/u)fsu* / **fisi*, *yifsa*	**fasaw*, *yafsuw*	—	*fisi*, *yifsi*
heda, *jehda* ‘to stop’	**hada*, *yihda*	**hadaʔ*, *yahdaʔ*	—	*hidi*, *yihda*
ġera, *jiġri* ‘to run’	**garaʕ*/*giri*, *yagri*	**garay*, *yagriy*	*žra*, *yižri*	*giri*, *yigri*
kera, *jikri* ‘to rent’	**kara*, *y(i/u)kri*	**ʔakray*, *yukriy*	*kra*, *yikri*	*kara*, *yikri*
kesa, *jiksi* ‘to cover’	**kasa*, *yaksu*	**kasaw*, *yaksuw*	*ksa*, *yiksi*	*kasa*, *yiksi*
kewa, *jikwi* ‘to burn’	**kawa*, *yakwi*	**kaway*, *yakwiy*	*kwa*, *yikwi*	*kawa*, *yikwi*
lema, *jilma* ‘to flash’	**lamaʕ*, *yilmaʕ*	**lamaʕ*, *yalmaʕ*	*lmaʕ*, *yilmaʕ*	*lamaʕ*, *yilmaʕ*
lewa, *jilwi* ‘to bend’	**lawa*, *yalwi*	**laway*, *yalwiy*	—	*lawa*, *yilwi*
mela, *jimla* ‘to fill’	**mala*, *yimla*	**malaʔ*, *yamlaʔ*	*mla*, *yimla*	*mala*, *yimla*
mexa, *jimxi* ‘to walk’	**mašaʕ*/*miši*, *yamši*	**mašay*, *yamšiy*	*mša*, *yimši*	*miši*, *yimši*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
neka, *jinki* ‘to tease’	**naka*, *yanki*	**nakay*, *yankiy*	—	—
nesa, *jinsa* ‘to forget’	**nisi*, *yinsa*	**nasiy*, *yinsay*	*nsa*, *yinsa*	*nisi*, *yinsa*
sewa, *jiswa* ‘to cost’	**siwi*/*sawa*, *yiswa*	**sawiy*, *yisway*	*swa*, *yiswa*	*sawa*, *yiswa*
tena, *jitni* ‘to fold’	**θana*, *yaθni*	**θanay*, *yaθniy*	*θna*, *yiθni*	*tana*, *yitni*
xewa, *jixwi* ‘to roast’	**šawa*, *yašwi*	**šaway*, *yašwiy*	*šwa*, *yišwi*	*šawa*, *yišwi*
żena, *jiżni* ‘to fornicate’	**zana*, *yazni*	**zanay*, *yazniy*	—	*zana*, *yizni*

Table 13 shows all the final-weak verbs that (at least optionally) have CeCɑ Perfects, despite having a guttural or emphatic R1 or R2. In parallel with what we have seen for the hollow and geminate verbs, in final-weak verbs we see that an association has developed between /i/ as the final vowel in the Imperfect, and CeCɑ rather than CɑCɑ as the Perfect pattern. Thus, all but one of the verbs in Table 13 have jVCCi Imperfects and have therefore extended the association between Imperfects of this type and CeCɑ (rather than CɑCɑ) Perfects. The one exception is *safa*/*sefa*, *jisfa* ‘to clear up (sky)’, where there is no obvious motivation for the innovation of the CeCɑ Perfect pattern, given the /ɑ/ in the Imperfect, though it is conceivable that in the past there was a *jisfi* variant of the Imperfect, the result of levelling of the more conservative Early Urban Arabic Imperfect pattern **y(i/u)ṣfu*.

Table 13: Final-weak verbs with /e/ in the Perfect despite presence of guttural or emphatic

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
rama/*rema*, *jarmi* ‘to throw away’	**rama*, *yarmi*	**ramay*, *yarmiy*	*ṛma*, *yaṛma*	*rama*, *yirmi*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
għala/*għela*, *jagħli* 'to boil'	**ɣala*/*ɣili*, *yaɣli*	**ɣalay*, *yaɣliy*	*ɣla*, *yaɣli*	*ɣili*, *yiɣli*
(*ħaba*/)*ħeba*, *jaħbi* 'to hide'	**ḫaba*, *yaḫba*/*yaḫbi*	**ḫabaʔ*, *yaḫbaʔ*	—	—
ħela, *jaħli* 'to waste'	**ḫala*, *yaḫli*	**ʔaḫlaw*, *yuḫliy*	*ḫla*, *yaḫli*	*ḫala*, *yiḫli*
(*ħala*/)*ħela*, *jaħli* 'to become sweet'	**ḥala*, *y(i/u)ḥlu* / *ḥili*, *yiḥla*	**ḥaluw* *yaḥluw* / *ḥaliy*, *yiḥlaw*	—	*ḥili*, *yiḥla*
ħema, *jaħmi* 'to heat'	**ḥama*, *yaḥmi*	**ʔaḥmaw*, *yuḥmi*	—	*ḥama*, *yiḥmi*
ħesa, *jaħsi* 'to castrate'	**ḫaṣa*, *yaḫṣi*	**ḫaṣay*, *yaḫṣi*	*ḫṣa*, *yaḫṣi*	*ḫaṣa*, *yiḫṣi*
ħexa, *jaħxi* 'to have sex'	**ḥaša*, *yaḥši*	**ḥašaw*, *yaḥšiy*	*ḥša*, *yaḥši*	*ḥaša*, *yiḥši*
qeda, *jaqdi* 'to serve'	**qaḍa*, *yaqḍi*	**qaʟʕay*, *yaqʟʕiy*	*qḍa*, *yaqḍi*	*ʔaḍa*, *yiʔḍi*
qela, *jaqli* 'to fry'	**qala*, *yaqli*	**qalaw*, *yaqluw* / **qalay*, *yaqliy*	*qla*, *yaqli*	*ʔala*, *yiʔli*
reħa, *jerħi*/*jirħi* 'to let go'	**raḫa*, *yarḫi*	**ʔarḫaw*, *yurḫiy*	—	*raḫa*, *yirḫi*
safa/*sefa*, *jisfa* 'to clear up (sky)'	**ṣafa*, *y(i/u)ṣfu* / *ṣifi*, *yiṣfa*	**ṣafaw* *yaṣfuw*	—	*ṣifi*, *yiṣfa*
saqa/*seqa*, *jisqi* 'to water'	**saqa*, *yasqi*	**saqay*, *yasqi*	*sqa*, *yisqi*	*saʔa*, *yisʔi*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
tafal tefa, *jitfi* ‘to extinguish’	**ṭafa*, *yaṭfi*	**ʔaṭfaʔ*, *yuṭfiʔ*	*ṭfa*, *yiṭfi*	*ṭafa*, *yiṭfi*
tewa, *jitwi* ‘to wrap’	**ṭawa*, *yaṭwi*	**ṭaway*, *yaṭwiy*	*ṭwa*, *yiṭwi*	*ṭawa*, *yiṭwi*

There are three further final-weak verbs with a quite different pattern. These are shown in Table 14.

Table 14: /o/-type final-weak verbs

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
għola/*għala*, *jogħla* ‘to become expensive’	**ɣala*, *y(i/u)ɣlu* / *ɣili*, *yiɣla*	**ɣalaw*, *yaɣluw*	—	*ɣili*, *yiɣla*
għola, *jogħla* ‘to rise’	**ʕala*, *y(i/u)ʕlu* / *ʕili*, *yiʕla*	**ʕalaw*, *yaʕluw* / *ʕaliy*, *yiʕlaw*	—	*ʕili*, *yiʕla*
għoxa, *jogħxa* ‘to faint’	**ɣuši*, *y(i/u)ɣša* / *ɣiši*, *yiɣša*	**ɣušiy*, *yuɣšaw*	—	*ɣiši*, *yiɣša*

Of these three, *għoxa*, *jogħxa* ‘to faint’ is the most straightforward. This is the regular reflex of a Proto-Arabic and Early Urban Arabic passive: **ɣuši*, *y(i/u)ɣša* (< **ɣušiy*, *yuɣšaw*). The active **ɣašiy* means ‘to cover’ and the passive **ɣušiy ʕalayh* means ‘to faint’ (literally ‘to be covered’). As noted in Lucas, Spagnol & Brown (2025), there are also two Maltese sound verbs which reflect original passive forms: *ħoloq*, *joħloq*, which formerly meant ‘to be born’ (Aquilina 1987) (< **ḫuluq*, *y(i/u)ḫluq* < **ḫuliq*, *yuḫlaq* ‘to be created’) and *għolob*, *jogħlob* ‘to become thin’ (< **ɣulub*, *y(i/u)ɣlub* < **ɣulib*, *yuɣlab* ‘to be defeated’).

The other two verbs in Table 14 cannot be explained in this way, but instead seem to be the result of an analogical extension to the verbs of the /o/ that is characteristic of several of the nominal and adjectival forms of these roots: *għolja* 'hill', *għoli* 'high', *ogħla* 'higher, more expensive'.[15]

6 Initial-weak verbs

The initial-weak verbs in Maltese are those in which R1 is **w*. Evidence for the reconstruction of **w*-initial verbs in Proto-Arabic is somewhat scarce (see Al-Jallad 2015: 119; Huehnergard 2019: 66), but what evidence there is speaks in favour of a system similar to that found in MSA, whereby they behave regularly in the Perfect, but the initial weak radical drops in the Imperfect of many but not all verbs, as in **waṣal*, *yaṣil* 'to arrive', but **wagiʕ*, *yīgaʕ* 'to ache' (underlyingly **yiwgaʕ*).

There does not appear to be a clear phonetic motivation for the dropping of R1 in some **w*-initial verbs and its retention in others, but there is a certain amount of regularity to the distribution, which is linked to the relationship between stem vowels in the Perfect and Imperfect. In order to understand this relationship, we need to revisit the Barth-Ginsberg alternation discussed in Section 5. For Northwest Semitic languages, this alternation is typically stated in terms of a relationship between the stem vowel and the prefix vowel in the Imperfect. That is, if the stem vowel is **a*, then the prefix vowel is **i*, as in **yirkab* 'to ride', otherwise the prefix vowel is **a*, as in **yaktub* 'to write' and *yaʔsir* 'to take prisoner'. This statement fails to fully account for the Proto-Arabic facts, however, as here we have a large class of verbs that have a guttural consonant for R2 or R3 and, as a consequence, have **a* as the stem vowel in the Imperfect in place of original **i*, but also **a* (not **i*) as the prefix vowel, as in **baʕaθ*, *yabʕaθ* 'to send'.[16] In view of the existence of verbs of this kind, a more accurate generalization regarding vowelling in Proto-Arabic is that the Imperfect prefix vowel is **i* only with verbs that have the form *CɑCiC in the Perfect, as in **rakib*, *yirkab*. This generalization captures the data for all Proto-Arabic strong verbs, in which an **i* stem vowel in the Perfect always corresponds to an **a* stem vowel in the Imperfect (*CɑCiC ↔ *yiCCɑC).

Returning to the **w-*initial verbs, however, here we do not find the same Perfect–Imperfect correspondence. For Classical Arabic, the grammarians record a number of verbs in which R1 is dropped in the Imperfect and /i/ is the stem vowel in both the Perfect and the Imperfect (e.g. Wright 1874: 78), as in *waθiq*, *yaθiq* 'to trust' and *wariθ*, *yariθ* 'to inherit'. It is not clear why these verbs defy the basic *CɑCiC ↔ *yiCCɑC pattern, especially since a number of other *wɑCiC verbs conform to it, as in the aforementioned **wagiʕ*, *yīgaʕ* 'to ache' (underlyingly **yiwgaʕ*), and also **waḥil*, *yīḥal* 'to become stuck' (underlyingly **yiwḥal*), both of which retain R1 in the Imperfect. In addition, there are two verbs, which in Classical Arabic have wɑCiC Perfects and the expected /ɑ/ stem vowel in the Imperfect, but which nevertheless drop R1 in the Imperfect: *waṭiʔ*, *yaṭaʔ* 'to tread' and *wasiʕ*, *yasaʕ* 'to be wide', which can presumably be reconstructed as such for Proto-Arabic. It remains to be seen whether a satisfactory account can be found for how this heterogeneous state of affairs came about. But to summarize, we can say that, except for *waṭiʔ* and *wasiʕ*, if a **w-*initial verb has **i* as the Perfect stem vowel and **a* as the Imperfect stem vowel, then it will retain R1 in the Imperfect and have **i* as the prefix vowel. Other **w-*initial verbs drop R1 in the Imperfect[17] and have **a* as the prefix vowel, as in **waṣal*, *yaṣil* 'to arrive', and **waqaʕ*, *yaqaʕ* 'to fall'.

A superficial look at the contemporary Arabic dialects might encourage a view that these represent a more conservative stage of **w-*initial verbs than we find in Classical Arabic, namely one where R1 was retained across the board. In Cairene, for example, R1 is retained in the Imperfect of verbs that drop it in Classical Arabic, as in *wiṣil*, *yiwṣal* 'to arrive', and the Tunisian and Levantine Imperfects *yūṣil* and *yūṣal* look like regular developments from the Cairene pattern, with the sound change **iw* > /ū/. On closer inspection, however, *yū-*/*yiw-* Imperfect prefixes with such verbs look more like an innovation. In much of the Levant, such verbs have an alternative Imperfect form with a short prefix vowel: *yəṣal* (Cowell 1964: 74), and the two Cairene verbs *wiʔif* 'to stop' and *wiʔiʕ* 'to fall' also have Imperfects with a short vowel: *yuʔaf*, *yuʔaʕ* (Woidich 2006: 81).

Against this background, Maltese exhibits notable conservatism among these verbs. As can be seen in Tables 15–16, with the single exception of *waqaf*, *jieqaf* 'to stop' (discussed below), wherever Proto-Arabic drops R1 in the Imperfect and has a short prefix vowel, this is maintained in Maltese, whereas Tunisian and Cairene have a long

vowel or a diphthong in every case (at least optionally). The extension of *yū-*/*yiw-* Imperfects to all **w*-initial verbs must therefore be a post-Early Urban Arabic innovation.

The Maltese initial-weak verbs that have developed regularly from Early Urban Arabic, following the sounds laws set out in Section 2 (but also with expected levelling of the **ya-* prefix to **yi-* before non-gutturals), are shown in Table 15.

Table 15: Regularly developed initial-weak verbs

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
waqa', *jaqa'* 'to fall'	**waqaʕ*, *yaqaʕ*	**waqaʕ*, *yaqaʕ*	*wqaʕ*, *yāqaʕ*/*yūqaʕ*	*wiʔiʕ*/*waʔaʕ*, *yuʔaʕ* / *yūʔaʕ* / *yiwʔaʕ*
wiled, *jiled* 'to beget'	**walad*/*wilid*, *yalid*/*yilad*	**walad*, *yalid*	*wlid*, *yūlid*	*wilid*, *yiwlid*
wiret, *jiret* 'to inherit'	**wiriθ*, *yariθ*	**wiriθ*, *yariθ*	*wriθ*, *yūriθ*	(*waras*/*wiris*, *yiwris*)
wiżen, *jiżen* 'to weigh'	**wazan*/*wizin*, *yizan*/*yazin*	**wazan*, *yazin*	*wzin*, *yūzin*	*wazan*, *yiwzin*

Table 16: Irregularly developed initial-weak verbs

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
waqaf, *jieqaf* 'to stop'	**waqaf*, *yaqaf*	**waqaf*, *yaqif*	*wqif*, *yāqif*	*wiʔif*, *yuʔaf* / *yūʔaf* / *yiwʔaf*
wasal, *jasal* 'to arrive'	**waṣal*/*wiṣil*, *yaṣil*/*yaṣal*	**waṣal*, *yaṣil*	*wṣil*, *yūṣil*	*wiṣil*/*waṣal*, *yiwṣal*
weġa'/*waġa'*, *jūġa'* 'to ache'	**wagaʕ*, *yiwgaʕ*	**wagiʕ*, *yiwgaʕ*	*wžaʕ*, *yūžaʕ*	*wagaʕ*, *yiwgaʕ*
weħel, *jeħel* 'to get stuck'	**wiḥil*, *yiwḥal*	**wiḥil*, *yiwḥal*	*wḥil*, *yūḥal*	*wiḥil*, *yiwḥil*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
wesa'/wasa', *jesa'/jasa'* 'to contain'	**wisiʕ*, *yasaʕ*	**wasiʕ*, *yasaʕ*	*wsaʕ*, *yūsaʕ*	*wisiʕ*, *yiwsaʕ*

The verbs shown in Table 16 all exhibit at least some irregularity in their historical development. The Perfects of *wasal* 'to arrive', *weħel* 'to get stuck', and *waqaf* 'to stop' are all regular. The expected Imperfect of *wasal* is ***jisal*, since there is no guttural to prevent levelling of the prefix vowel to /i/. The attested form *jasal* is to be explained as the result of pressure to create vowel harmony between the stem vowel and prefix vowel, as well as parallelism between the vowelling of the Perfect and Imperfect (see Lucas, Spagnol & Brown 2025 for numerous vowel-harmonic changes of this kind in the strong verbs). The same is true for *jeħel*. Expected here would be ***jiwħel* (see below for discussion of *juġa'* 'to ache'). The anomalous long vowel in *jieqaf* 'to stop' represents an innovation in the immediate common ancestor of Maltese and Tunisian (**yaqif* > **yāqif*). This is evidently an analogical extension of the **yākul* and **yāḫuð* Imperfects discussed below in Section 7. Note that Tunisian has further extended this *yā-* prefix to its reflex of the verb for 'to fall': *wqaʕ*, *yāqaʕ*. The stem vowel /ɑ/ in *jieqaf* is also not the regularly expected /e/ (compare Tunisian *yāqif*, which shares with Maltese the rule of post-stress neutralization of **a* and **i*). This presumably represents the outcome of pressure for matching vowels after R2 in both the Perfect and Imperfect.

The two remaining verbs in Table 16 have variant Perfect forms: *weġa'/waġa'*, 'to ache' and *wesa'/wasa'* 'to contain'. The form *weġa'* is the regularly expected form of Early Urban Arabic **wagaʕ*. The variant *waġa'* must be the result of copying back the vowel of the second syllable to the first in order to re-establish vowel harmony. As shown by Lucas, Spagnol & Brown (2025), this a commonly observed phenomenon in sound verbs with a guttural R3 (e.g. *bela'/bala'*, *nefaħ/nafaħ*, *neża'/naża'*, *żera'/żara'*, etc.). This must also be the explanation of the variant form *wasa'* from *wesa'*, but *wesa'* itself is also not the regularly expected reflex of *wisiʕ*, which would be *wisaʕ*. However, there are no CiCɑC verbs in Maltese. This is because of a far-reaching process, detailed by Lucas, Spagnol & Brown (2025), of re-establishing vowel harmony in Perfects in the immediate common ancestor of Tunisian and Maltese, after the disruption caused by

the post-stress neutralization rule. Thus we had, for example with the verb ‘to be happy’, **firiḥ* > **firaḥ* > **faraḥ*, the regular outcome of which in Maltese is *feraħ*, and exactly parallel to this we have **wisiʕ* > **wisaʕ* > **wasaʕ* > Maltese *wesa’*.

The two variant Imperfect forms for ‘to contain’—*jesa’* and *jasa’* instead of regularly expected ***jisa’*—are transparently the result of establishing parallelism with the respective Perfect forms *wesa’* and *wasa’*. The form *juġa’* (IPA /ˈjuːdʒɐ/) ‘to ache’ is noteworthy due to its long /ū/ in the prefix.[18] This reflects the retained R1 in the reconstructed form **yiwgaʕ*. Given that /iw/ is a licit (albeit rare) sequence syllable-internally in contemporary Maltese (as in the nouns of instance *tiwja* ‘a fold’ and *xiwja* ‘a roasting’), it is somewhat surprising that the same sequence is reflected as /ū/ in *juġa’*. But with only this single instance of an initial-weak verb with retained R1 in Maltese, and so much analogical change in this domain in other descendants of Early Urban Arabic, it does not seem possible to account for this development at present.

7 Miscellaneous irregular verbs

In this final section we consider the nine verbs in Table 17, which are both weak and irregular.

Table 17: Miscellaneous irregular verbs

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
ġie, *jiġi* ‘to come’	**gā*, *yagi*	**gayaʔ*, *yagyiʔ*	*ža*, *yži*	ga/*gih*, *yīgi*
ħa, *jieħu* ‘to take’	**ʔaḫað*, *yāḫuð*	**ʔaḫað*, *yaʔḫuð*	*ḫða*, *yāḫu(ð)*	*ḫad*, *yāḫud*
kiel, *jiekol* ‘to eat’	**ʔakal*, *yākul*	**ʔakal*, *yaʔkul*	*kla*, *yākul*	*kal*, *yākul*
mar, *imur* ‘to go’	**marr*, *y(i/u)murr*	**marr*, *yamurr*	—	*marr*, *yumurr*

Maltese	Early Urban Arabic	Proto-Arabic	Tunisian	Cairene
(*għad*,) *jgħid* ‘to say’	**ʕād*, *yaʕīd*	**ʔaʕwad*, *yuʕwid*	—	*ʕād*, *yiʕīd*
(*għaraf*,) *jaf* ‘to know’	**ʕaṛaf*, *yaʕṛif*	**ʕaraf*, *yaʕrif*	*ʕṛaf*, *yaʕṛaf*	*ʕirif*, *yiʕṛaf*
qal, (*iqul*) ‘to say’	**qāl*, *y(i/u)qūl*	**qawal*, *yaqwul*	*qāl*, *yqūl*	*ʔāl*, *yuqūl*
ra, *jara* ‘to see’	**rā*, *yara*	*raʔay*, *yarʔay*	*ra*, *yra*	(*raʔa*, *yara*)[19]
ta, *jagħti* ‘to give’	**ʕaṭa*, *yaʕṭi*	*ʕaṭay*, *yaʕṭiy*	—	*ʕaṭa*, *yiʕṭi*

As is well known, the Maltese verb ‘to say’ is suppletive. In the Imperfect and first and second person of the Perfect, the forms go back to Proto-Arabic **ʔaʕwad*, *yuʕwid* ‘to cause to return’, while the third-person Perfect forms go back to **qawal*, *yaqūl* ‘to say’. The Maltese reflexes of these two hollow verbs are the regularly expected outcomes of these proto-forms.

The verb *mar*, *imur* ‘to go’ perhaps also represents a combination of two distinct Proto-Arabic verbs. The Perfect forms *marret* ‘she went’, *marru* ‘they went’, and the plural Imperfect forms such as *imorru* ‘they go’, clearly indicate that we are dealing with a reflex of the geminate verb **marr*, *yamurr* ‘to pass’. The other forms, such as *mar* ‘he went’, *mort* ‘I went’ and *imur* ‘he goes’ look exactly like a hollow verb. Either this is a highly irregular and apparently unmotivated partial conversion of a geminate verb into a hollow verb, or, perhaps more likely (and as Aquilina 1990 suggests), these other forms reflect the Proto-Arabic verb **mawar*, *yamwur* ‘to move to and fro’.

Regarding ‘to give’, here the Imperfect form *jagħti* is the regular reflex of Early Urban Arabic **yaʕṭi*. In the Perfect, the expected reflex of **ʕaṭa* would be ***għata*, but we have irregular loss of R1 (which must predate the generalized loss of **ʕ* in standard Maltese), leaving only *ta*. We have a similar development with *jaf* ‘to know’, except that this lacks a Perfect form in contemporary Maltese. The regular reflex of the Early

Urban Arabic form **yaʕṛif* would be *jagħraf*. This is of course attested, but functions as a distinct verb *għaraf*, *jagħraf* 'to recognize'. It is only in this highly reduced form, with irregular loss of both R1 and R2, that it means 'to know'.

The remaining verbs in Table 17 are all irregular on account of being derived from Proto-Arabic weak verbs which additionally had a glottal stop for one of the radicals, the latter being lost in Early Urban Arabic, except in initial position, as in the case of the Perfect forms **ʔakal* 'to eat' and **ʔaḫað* 'to take'. Beginning with 'to eat', the Maltese Imperfect form *jiekol* is the regular reflex of Early Urban Arabic **yākul*, itself the regular reflex of Proto-Arabic *yaʔkul*, with loss of the glottal stop and compensatory lengthening of the prefix vowel. Some Arabic varieties, such as Levantine, maintain the initial glottal stop in the Perfect of both **ʔakal* and **ʔaḫað*. Others, such as Cairene, simply delete it, giving *kal* and *ḫad*. This, however, results in a verb with only two radicals, giving rise to pressure to re-establish triradicality. In the case of these two verbs, Cairene resists this pressure. In Tunisian, however, both verbs are reconfigured as final-weak: *kla* and *ḫða*. In Maltese 'to eat' is reconfigured in the Perfect as hollow: *kiel*. With 'to take', Maltese not only resists the pressure to re-establish triradicality, it also deletes the original R3, leaving only *ħa* for the third person singular masculine, though R3 is retained in all other parts of the Perfect paradigm. R3 is also deleted in the Imperfect singular, but not the plural, giving *jieħu* 'he takes' but *jieħdu* 'they take'.

The verb 'to come' derives from Proto-Arabic **gayaʔ*, *yagyiʔ*, a root that is both hollow and has a glottal stop for R3. This develops regularly into Early Urban Arabic **gā*, *yagi*. The Maltese Perfect form *ġie* is the regular reflex of **gā*. The Imperfect form *jiġi* (IPA /ˈjidʒɪ/) is the expected development of **yagi*, with levelling of the prefix vowel to /i/. The pressure to restore triradicality is therefore resisted here (in standard Maltese at least), unlike, for example, in Cairene, where the prefix vowel is lengthened: *yīgi*.

Finally, *ra* 'to see' is parallel to *ġie*, except that here we have final-weak verb with a glottal stop as the original R2: **raʔay*. The Maltese Perfect form *ra* is the regular reflex of Early Urban Arabic **rā*. The Imperfect form *jara* (IPA /ˈjɐːrɐ/), however, is irregular in having a lengthened prefix vowel, presumably due both to pressure to restore triradicality, and analogy with *jieħu*, *jiekol* and *jieqaf*.

8 Conclusion

In this article we have shown that the considerable complexity and heterogeneity exhibited by Maltese weak verbs is not the result of random change, but can be straightforwardly accounted for on the basis of properly established exceptionless sound laws, together with the operation of simple analogical processes with generally obvious motivation. Another important point to highlight is that while we have seen numerous examples here of how Maltese has innovated, relative to the Arabic dialects, we should not overlook the fact that many areas of Maltese grammar exhibit considerable conservatism, having failed to participate in innovations that have spread through the Arabic dialects in the eight centuries since Maltese split from North African Arabic. The clearest example of this conservatism that we have seen in the present article concerns the maintenance of short vowels in the prefixes of initial-weak verbs, discussed in Section 6.

Much more work is required before we have a complete understanding of the historical development of Maltese—in all its varieties—from its Proto-Arabic origins. With the present article and Lucas, Spagnol & Brown (2025) we have dealt with the Form I verbs, but many other areas of morphophonology remain for the Maltese linguistics community to investigate, including the derived verb forms, participles, verbal nouns, adjectives, and more.

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1. Unless otherwise indicated, “Maltese” should be understood here as referring to contemporary standard Maltese.↩

2. We are grateful to Marijn van Putten and Lameen Souag for their exceptionally helpful and constructive comments on an earlier draft of this article. Any remaining errors are our own.↩

3. Available at: https://mlrs.research.um.edu.mt. Accessed 26/03/2024.↩

4. See, for example, the entries for تاقة in the crowdsourced Arabic dialect dictionary mo3jam.com: https://ar.mo3jam.com/term/تاقة (https://ar.mo3jam.com/term/%D8%AA%D8%A7%D9%82%D8%A9), accessed 26/03/2025.↩

5. In these Imperfect verb forms we observe the so-called Barth-Ginsberg alternation, which for Proto-Arabic is best stated as follows. Verbs with CɑCiC Perfects have **a* as the stem vowel and **i* as the Prefix vowel in the Imperfect (but see Section 6 for some exceptions); with all other verbs the prefix vowel is **a*. As discussed by van Putten (2020) and Lucas, Spagnol & Brown (2025), the alternation is levelled in favour of a prefix vowel of /ɑ/ across the board in the variety of Classical Arabic that became MSA, and, in many Arabic dialects, is levelled across the board in favour of /i/ as the prefix vowel (often harmonizing with a /u/ stem vowel). However, traces of the Barth-Ginsberg alternation survive in Maltese. Specifically, in roots where R1 is a guttural consonant, an inherited /ɑ/ as the prefix vowel is preserved. A variation on this pattern can be observed in the Imperfect forms of several of the final-weak verbs discussed in Section 5.↩

6. The default vowel of the Imperfect prefix in Cairene is /i/. Where the stem vowel is /u/ or /ū/, the prefix vowel is optionally realized as /u/. For simplicity, we represent it as /u/ here whenever the stem vowel is /u/ or /ū/.↩

7. In Classical Arabic, the meaning of ‘to recover, wake up’ is restricted to Form IV of this root. Form IV merged with Form I in Early Urban Arabic. It appears that Cairene preserves a reflex of the original Form I Imperfect (*yufūʔ* < **yafūq*), while Maltese and Tunisian preserve a reflex of the original Form IV Imperfect (*ifiq* < **yufīq*).↩

8. Dictionaries of Classical Arabic (e.g. Lane 1863: 2314–5) cite two distinct verbs: *ɣār*, *yaɣār* ‘to envy’ and *ɣār*, *yaɣīr* ‘to benefit’. Either the two verbs were conflated in Early Urban Arabic with only the ‘envy’ meaning surviving, or **ɣār*, *yiɣār* ‘to envy’ innovated a new Imperfect **yiɣīr*. Seeger (2022: ĠYR) lists both *yɣīr* and *yɣār* as possible Imperfect forms of this verb in Palestinian Arabic.↩

9. In Cairene, **q* is reflected as /ʔ/ in inherited items and does not block *imāla*. In reborrowings from Standard Arabic, /q/ occurs and behaves like an emphatic with respect to the blocking of *imāla*.↩

10. Inflectional suffixes are included in this example as they are relevant to the point at hand. Elsewhere in this article we omit inflectional suffixes in the third person, for ease of comparison between Proto-Arabic, which retains them, and other varieties which do not.↩

11. As noted in Section 2, while **r* does not generally function as emphatic in initial position, in a large subset of roots in Tunisian and other Maghrebi Arabic varieties, it has been reanalysed as phonemically emphatic in all contexts. This is the case with both *ṛadd* ‘to give back’ and *ṛašš* ‘to spray’ in Tunisian (Singer 1984: 352). We can therefore assume that both these items were emphatic in earlier Maltese.↩

12. This verb exhibits an irregular but (for Maltese) not uncommon fortition of initial **h* to **ḥ*.↩

13. In contemporary standard Maltese, this verb is typically used in the Imperfect and the Imperative. As Perfect forms, Aquilina (1990) cites both the regularly expected form *seff*, and alternative form *saff*.↩

14. Note that several Early Urban Arabic (and therefore Maltese) final-weak verbs derive from Proto-Arabic strong verbs with **ʔ* as R3, which is lost essentially across the board in Arabic dialects (e.g. *qara*, *jaqra* ‘to read’ < **qaraʔ*, *yaqraʔ*). Maltese has also gained a final-weak verb in *lema*, *jilma* ‘to flash’ (< **lama*, *yalmaʕ*)

via irregular loss of final *ʕ. Conversely, *baqa’*, *jibqa’* is treated in the third person of the Perfect, but not the first and second persons, as if it were derived from a root with *ʕ as R3, when in reality it is derived from a final-weak root √BQY.↩

15. It must be noted, however, that the /o/ in these forms also cannot be explained in terms of regular sound laws. A full explanation of the provenance of this /o/ is beyond our present focus, but it seems likely that there is an analogical link with the irregular comparative forms *oħla* ‘sweeter’ and *ogħna* ‘richer’, themselves apparently modelled on the regularly developed feminine form *oħra* ‘other’ (< *ʔuḫrā*).↩

16. The behaviour of verbs of this type is clearly set out by al-Farrāʔ (2014: 8–9), in his comments on Quranic reading traditions that exhibit the Barth-Ginsberg alternation (i.e. traditions which, unlike the variety of Classical Arabic which became MSA, have not levelled the Imperfect prefix vowel to /ɑ/ across the board). Al-Farrāʔ cites *qaraʔ*, *yaqraʔ* ‘to read’ and *ðahab*, *yaðhab* ‘to go’ as examples of this class of verb.↩

17. For completeness, we should note that there are also a small number of *wɑCuC, yɑwCuC verbs, which have the expected **a* prefix vowel, but which also retain R1 in the Imperfect, as in **wabuʔ*, *yawbuʔ* ‘to be plagued’ (see Wright 1874: 79). None of these survive into Maltese, however.↩

18. This verb almost always takes a pronoun suffix, as in *dahri juġagħni* ‘my back hurts’, which shifts stress to the penultimate syllable, making the length of the first vowel hard to discern. It is clearly long when not suffixed, however, as in the proverb *għajn ma tara*, *qalb ma tuġa’* ‘what the eye doesn’t see, the heart can’t grieve over’.↩

19. The Cairene verb *raʔa*, *yara* ‘to see in a vision’ is better understood as a reborrowing from Classical Arabic than as the direct outcome of inheritance from Early Urban Arabic, as with the Maltese and Tunisian reflexes.↩